GeoAI Console: Land Use Classification & Temporal Change Detection using Deep Learning

Celebal Technologies Internship Project Report
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1. Abstract

This report details the development of the GeoAI Console, a deep learning-based system designed for satellite image land-use classification and temporal change detection. The project leverages Transfer Learning via ResNet18 to classify aerial imagery and utilizes 512-dimensional feature embeddings coupled with cosine similarity to identify significant land-use changes over time. Developed using PyTorch and deployed via an interactive Flask dashboard, the system is trained on the EuroSAT RGB dataset and evaluated for generalizability using the UC Merced Land Use Dataset.

2. Introduction

The rapid expansion of urban areas, deforestation, and agricultural shifts necessitates continuous monitoring of land surface changes. Geographic Artificial Intelligence (GeoAI) integrates geospatial data with deep learning architectures to automate the extraction of actionable intelligence from satellite imagery. This project, completed during the Celebal Technologies internship, establishes an end-to-end pipeline capable of categorizing distinct land-use types and quantifying temporal shifts between chronological satellite image pairs, thereby offering a scalable alternative to manual GIS analysis.

3. Problem Statement

Traditional land-use monitoring relies on manual visual inspection of satellite imagery, which is resource-intensive, highly subjective, and unscalable across large geographic territories. Furthermore, detecting temporal changes requires pixel-perfect alignment and complex spectral analysis. There is a critical industry requirement for an automated, deep learning-driven framework that can accurately classify land use from standard RGB imagery and automatically flag regions that have undergone structural or categorical transformations over time.

4. Objectives

- Engineer a robust deep learning pipeline for multi-class land-use classification.
- Implement a reliable temporal change detection mechanism using feature extraction and similarity metrics.
- Evaluate the efficacy of Transfer Learning by comparing frozen and fine-tuned ResNet18 architectures against a baseline Convolutional Neural Network (CNN).
- Develop an interactive, web-based Flask dashboard to visualize datasets, training metrics, and change detection heatmaps.

5. Dataset Description

The primary corpus for model training and internal validation is the EuroSAT RGB Dataset, comprising 27,000 georeferenced image patches sampled from Sentinel-2 satellite data. The dataset spans 10 distinct land-use classes, including Annual Crop, Forest, Highway, Industrial, and Residential areas.

To assess cross-domain generalizability, the UC Merced Land Use Dataset is utilized exclusively as an external evaluation benchmark, ensuring the model's feature extraction capabilities remain robust outside the EuroSAT distribution.

6. Data Pipeline

The data pipeline encompasses stringent preprocessing protocols to optimize the imagery for deep learning consumption. Images are originally 64x64 but are resized to 224x224 pixels and normalized utilizing ImageNet mean and standard deviation parameters to align with the pre-trained ResNet18 input requirements. Data augmentation techniques, including random horizontal flips, random rotations, and color jittering, are applied to the training subset to prevent overfitting.

The dataset is partitioned into training (70% - 18,900 images), validation (15% - 4,050 images), and testing (15% - 4,050 images) subsets following a stratified random split to maintain uniform class distribution. It is critical to note that the EuroSAT RGB dataset lacks embedded spatial metadata (exact geographic coordinates). Consequently, a reproducible random split utilizing fixed random seeds was implemented. This methodology introduces a known limitation regarding spatial leakage, as overlapping or adjacent geographical patches may inadvertently be distributed across both the training and testing sets.

7. Model Development

- **Baseline CNN:** A custom 3-layer Convolutional Neural Network was constructed to serve as a benchmark for computational efficiency and baseline accuracy.
- **Transfer Learning (ResNet18 - Frozen):** A pre-trained ResNet18 architecture was instantiated with its convolutional feature extraction layers (Layer1, Layer2) frozen.

Only the final fully connected layer was modified and trained to output the 10 target classes, testing the efficacy of generic ImageNet features on aerial data.

- **Transfer Learning (ResNet18 - Fine-Tuned):** The ResNet18 network was partially unfrozen (Layer3, Layer4, and fully connected layer active, representing 11.2M trainable parameters), allowing gradient updates. This enabled the model to specialize its hierarchical feature maps specifically for remote sensing topography, significantly improving discriminative power.

8. Experimental Setup

The models were trained utilizing the PyTorch framework on CUDA devices. The primary hyperparameters and configuration details are documented below.

Configuration Parameter	Value
Optimizer	Adam
Loss Function	CrossEntropyLoss
Epochs	5
Batch Size	32
Learning Rate	0.0001 (1e-4)
Random Seed	42

9. Results

The performance of the fine-tuned ResNet18 model yielded superior classification metrics compared to the frozen architecture and baseline CNN. Comprehensive evaluation metrics are recorded below.

Metric	Value
Final Training Loss	0.1127
Final Validation Loss	0.0812
Test Accuracy (EuroSAT)	96.99%
Macro Precision	97.04%
Macro Recall	96.99%
Macro F1 Score	96.99%
External Accuracy (UC Merced)	95.87%

Model Comparison Table:

Architecture	Accuracy	Precision	Recall	F1 Score
ResNet18 (Fine-Tuned)	96.99%	97.04%	96.99%	96.99%
ResNet18 (Frozen)	88.00%	88.49%	88.00%	87.96%
Baseline CNN	74.89%	76.45%	74.89%	74.71%

Confusion Matrix Analysis:

Visualized across outputs located in outputs/resnet18_finetuned/confusion_matrix.png and outputs/uc_merced/confusion_matrix.png, showcasing minimal inter-class confusion, predominantly isolated to structurally similar vegetation or urban classes.

10. Change Detection

Temporal change detection is facilitated by leveraging the fine-tuned ResNet18 as a feature extractor. For a given spatial coordinate at time T1 and T2, the network drops the final classification head and outputs a 512-dimensional continuous feature embedding representing the localized spatial structure. The cosine similarity between the T1 and T2 embeddings is computed to quantify structural divergence (e.g., recorded sample output of 0.4819).

An empirical threshold of 0.6333, derived via Receiver Operating Characteristic (ROC) curve analysis (achieving an ROC AUC of 0.9889), determines the boundary between ambient variance (e.g., seasonal changes, illumination differences) and fundamental land-use alteration (e.g., deforestation, urbanization). Absolute difference heatmaps are generated to localize the pixel clusters contributing most significantly to the change.

- **Pair 1 (Forest to Forest):** Verified against outputs/change_detection/heatmaps/pair1_forest_forest/heatmap_overlay.png
- **Pair 2 (Forest to Residential):** Verified against outputs/change_detection/heatmaps/pair2_forest_residential/heatmap_overlay.png
- **Pair 3 (River to River):** Verified against outputs/change_detection/heatmaps/pair3_river_river/heatmap_overlay.png
- **Pair 4 (Industrial to SeaLake):** Verified against outputs/change_detection/heatmaps/pair4_industrial_sealake/heatmap_overlay.png
- **Pair 5 (Residential to Residential):** Verified against outputs/change_detection/heatmaps/pair5_residential_residential/heatmap_overlay.png

11. Dashboard

An interactive web application was constructed using Flask to abstract the underlying PyTorch pipeline into a user-friendly console. The application comprises the following modules:

- **Dashboard:** The central landing page providing an aggregated overview of system status and quick navigation links (Screenshots pending integration).
- **Dataset:** A visualization module allowing users to traverse class distributions and sample images from the EuroSAT corpus.
- **Training:** An interface exposing hyperparameter configuration options and real-time visualization of training/validation loss curves.

- **Evaluation:** Renders classification reports, ROC curves, and dynamic confusion matrices for the trained models.
- **Comparison:** A side-by-side analytical view contrasting the baseline CNN against frozen and fine-tuned ResNet architectures.
- **Change Detection:** The primary functional module where users can upload historical and current satellite image pairs to generate similarity scores and difference heatmaps.
- **About:** Details technical architecture, dataset citations, and internship context.

12. Error Analysis

Out of 4,050 test images, the model registered only 122 incorrect predictions, achieving an overall accuracy of 96.99%. A systematic review of the top 5 misclassified instances revealed specific pattern recognition limitations within the current architecture.

Rank	True Label	Predicted Label	Confidence	Failure Hypothesis
1	PermanentCrop	AnnualCrop	99.91%	Spectral and textural similarities between distinct crop stages over overlapping seasons.
2	Industrial	Highway	99.87%	Linear features, large concrete surfaces, and paved access roads mimicking highway structures.
3	PermanentCrop	AnnualCrop	99.83%	Ambiguous vegetation density lacking definitive structural differences at a 224x224 resolution.
4	Industrial	Residential	99.82%	Dense clustering of structures and similar roofing material reflectance profiles.
5	Industrial	Residential	99.70%	Close proximity

				of urban zones causing overlapping spatial features.

13. Spatial Leakage Discussion

In geospatial machine learning, partitioning data utilizing a true spatial block split—where data is segregated into distinct, non-overlapping geographic regions—is the industry standard to ensure robust model evaluation. A standard random split, by contrast, shuffles data arbitrarily, often placing contiguous satellite patches in both the training and test sets. This creates spatial leakage, as the model may memorize local geographical textures rather than learning generalized class features, artificially inflating validation metrics.

The EuroSAT RGB dataset, while extensive, is published without the precise spatial coordinate metadata required to perform a rigorous block split. Consequently, a reproducible random split was the only practical approach available for this implementation. This constraint is acknowledged, and the high evaluation metrics achieved on the internal test set must be interpreted with an understanding of potential spatial autocorrelation.

14. Limitations

- **Spectral Constraints:** The model operates exclusively on 3-channel RGB data, omitting valuable atmospheric and topological data present in Multi-Spectral (e.g., Near-Infrared) imagery.
- **Spatial Leakage:** As detailed in Section 13, the random data split limits guarantees on absolute geographical generalizability.
- **Threshold Sensitivity:** The ROC-derived threshold (0.6333) for change detection is highly sensitive to seasonal variations and may yield false positives during heavy cloud cover or extreme illumination shifts.

15. Future Improvements

- **Architectural Upgrades:** Transitioning from ResNet18 to more computationally efficient models like EfficientNet-B0 or state-of-the-art architectures such as Vision Transformers (ViT) to capture long-range spatial dependencies.
- **Explainability Integration:** Implementing Grad-CAM (Gradient-weighted Class Activation Mapping) to visually highlight the exact spatial regions influencing the model's classification decisions.
- **GIS Integration:** Interfacing the outputs directly with geographic information systems (GIS) platforms for automated mapping and coordinate tagging.

- **Multi-spectral Imagery:** Expanding the pipeline to ingest 13-channel Sentinel-2 data to leverage non-visible spectrum indices like NDVI (Normalized Difference Vegetation Index).
- **Cloud Deployment:** Containerizing the Flask console via Docker and deploying it on a scalable cloud infrastructure (AWS/Azure) for remote access and parallel processing capabilities.

16. Conclusion

The GeoAI Console fulfills the core objectives of automating land-use classification and temporal change detection using deep learning methodologies. The integration of fine-tuned ResNet18 feature extraction with cosine similarity analysis provides a highly functional, scalable framework for analyzing geographic alterations over time. While constrained by dataset metadata limitations regarding spatial leakage, external evaluation indicates robust feature extraction capabilities. The interactive Flask dashboard successfully abstracts the technical complexity, delivering a deployment-ready prototype aligned with modern industry requirements in the Remote Sensing domain.